

Mohamad Latifian

<https://www.latifian.me>

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Research Interests

My research lies at the intersection of theoretical computer science, artificial intelligence, and economics, with a focus on algorithms and computational complexity. My primary interests include computational social choice, fair division, algorithmic game theory, and mechanism design.

Education and Academic Positions

Postdoctoral Research Associate

University of Edinburgh, UK

Host: *Aris Filos-Ratsikas*

Jul. 2024 – present

Ph.D. in Computer Science

University of Toronto, Canada

Advisors: *Nisarg Shah and Allan Borodin*

Thesis: *Decoding Distortion: Pushing the Limits of Optimal Collective Decision-Making*

GPA: *4.0/4.0*

Sep. 2020 – Sep. 2024

M.Sc. in Algorithms and Computation

Sharif University of Technology, Iran

Advisor: *Mohammad Ghodsi*

Thesis: *Considering the Effect of Abstention in the Distortion Value.*

GPA: *19.59/20*

Sep. 2017 – Sep. 2019

B.Sc. in Computer Engineering

Sharif University of Technology, Iran

Advisor: *Hamid Zarrabi-Zadeh*

Thesis: *Hamiltonian Path in Solid Grid Graphs.*

GPA: *17.22/20*

Sep. 2012 – Jul. 2017

Conference Publications

(Authors in alphabetical order)

C18. .

C17. .

C16. .

C15. .

C14. .

C13. .

C12. .

C11. .

C10. .

C9. .

C8. .

C7. .

C6. .

C5. .

C4. .

C3. .

C2. .

C1. .

Journal Publications

J2. .

J1. .

Working Papers

(Authors in alphabetical order)

M6. .

(Under submission to WINE)

M5. .

M4. .

M3. .

M2. .

M1. .

Academic Visits

Long-Term

University of Oxford, UK.

Host: Edith Elkind

Sep. – Nov. 2023

Max Planck Institute for Informatics, Germany.

Host: Alkmini Sgouritsa

May – Jun. 2019

Work title: Resource-Aware Protocols for Network Cost-Sharing Games ([C-15.](#))

Short-Term

University of Oxford, UK.
Host: Alexandros Hollender

Sep. 2025

University of Edinburgh, UK.
Host: Aris Filos-Ratsikas

Nov. 2023

University of Essex, UK.
Host: Alexandros Voudouris

Oct. 2023

Professional Service

- **PC Member:** AAAI '27, EC '26, AAAI '26, EC '25, AAMAS '25, ICLR '25, AAAI '25, NeurIPS '24, IJCAI '24, AAMAS '24, AAAI '24, AAAI '21
- **Journal Reviewer:** Autonomous Agents and Multi-Agent Systems (JAAMAS), Artificial Intelligence (AIJ), Journal of Artificial Intelligence Research (JAIR)
- **Conference Reviewer:** SODA '27, ESA '26, STOC '25

Invited Talks

- T7. *Fair Division with Market Values*, Workshop on Fair Resource Allocation: Concepts, Algorithms and Complexity (FaiRACAC '25), held in conjunction with ICALP 2025 (7 Jul. 2025).
- T6. *Distortion in Social Choice Problems*, University of Edinburgh, EconCS Group Meetings (Oct. 2024).
- T5. *Fair Division with Market Values*, University of Edinburgh, Laboratory for Foundations of Computer Science (LFCS) Lab Lunch (Oct. 2024).
- T4. *How to Make Efficient (Collective) Decisions*, Tehran Institute for Advanced Studies (Sep. 2024).
- T3. *Best of Both Distortion Worlds*, University of Edinburgh, Laboratory for Foundations of Computer Science (LFCS) Seminar (Nov. 2023).
- T2. *Best of Both Distortion Worlds*, University of Essex, The Centre for Computational Finance and Economic Agents (CCFEA) Seminar Series (Oct. 2023).
- T1. *Resource-Aware Protocols for Network Cost-Sharing Games*, The Drexel EconCS Research Group Meetings (Jun. 2020).

Selected Teaching Experience**Guest Lecturer**

University of Edinburgh
Algorithms and Data Structures ([Course web page](#))

Oct. 2025

Teaching Assistant

University of Toronto

- Algorithm Design, Analysis & Complexity
- Data Structures and Analysis
- Algorithmic Game Theory and Mechanism Design

Fall '23

Winter '23

Fall '22

Teaching Assistant

Sharif University of Technology

(* := Lead TA)

- Algorithmic Game Theory
- Design and Analysis of Algorithms
- Theory of Languages and Automata

Fall '17, Spring '19*

Fall '14, Fall '18*

Fall '14, Spring '15, Fall '15, Spring '16

– Data Structures and Algorithms

Spring '15

Member of the Scientific Committee

Fall 2016

4th Sharif National Students Competition,
Sharif University of Technology.

Head of the Scientific Committee

Aug. 2013 and Aug. 2014

13th & 14th Mehregan Mathematics Workshop,
Shahid Ejeii High School.

Teaching Special Topics in Mathematics

2011–2013

Teaching high-school students preparing for Olympiad in Informatics.
Topics included Algorithms, Graph Theory, Problem-Solving Strategies, and Programming.

Supervision Experience

Mohammad Hossein Ghasemi and Mohammad Sharify (Co-supervised with A. Filos-Ratsikas)

Undergraduate students, Sharif University of Technology

Aug. 2025 – present

Title: *Metric Distortion with Graph Objectives*

Hamidreza Alipour, Kiarash Joolaei, Sepehr Mizanian

Undergraduate students, Sharif University of Technology

Nov. 2024 – present

Title: *A Deeper Look at Distortion in Impartial Culture Electorates*

Dana Afazeli

Undergraduate student, Sharif University of Technology

Aug. 2024 – Jan. 2026

Title: *Social Deliberation: Democratic Deliberation Through Unbiased Social Interactions* (Appeared in WWW '26)

Mehrad Abbaszadeh and Ali Ansarifar (Co-supervised with M. Seddighin)

Undergraduate students, Sharif University of Technology

Oct. 2023 – Aug. 2025

Title: *Metric Distortion with Preference Intensities* (Appeared in AAI '26)

Honors and Awards

- **C.C. Gotlieb (Kelly) Graduate Fellowship** in the Department of Computer Science, University of Toronto 2023
- **Silver medal** in the 21st Iranian National Olympiad in Informatics (INOI). 2011
- **Ranked 2nd and 7th** in Nationwide Graduate Entrance Exam, in Algorithms and Computation and Artificial Intelligence majors, respectively. 2017
- **Ranked 1st** in cumulative GPA among 97 M.Sc. students of the department, Sharif University of Technology. 2019
- Scholarship from Iranian National Elites Foundation 2012 – 2019

Activities

- **Elected Member & President** of the Students' Scientific Chapter (SSC) Computer Engineering Department of Sharif University of Technology. 2014 – 2015
- **Chief of Staff** of 17th & 18th International Collegiate Programming Contest (ICPC) Asia Region - Tehran Site. 2015 – 2016

- **Chief of Staff** of 6th & 7th AI Challenge

2014 – 2015

An event where AI agents written in Java, C++, or Python compete to pass the game field and overcome other agents.